Transfer Learning for
10 Monkey Species Image Classification

Project Report — Deep Learning using Keras

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Public Colab notebook:
Open notebook

Dataset: Kaggle — 10 Monkey Species

1. Overview

This project builds a ten-class image classifier using Keras transfer learning. MobileNetV2 pretrained on ImageNet serves as the feature extractor, with a newly added classification head. Six hyperparameter configurations were run. Model selection used validation accuracy; the highest recorded validation accuracy was 98.54%.

2. Dataset and split

The public Kaggle dataset was downloaded in Google Colab using kagglehub. The notebook pooled images from the source training and validation folders and created a new stratified split with random seed 42. The resulting dataset contained 1,369 images in ten class folders (n0–n9). Images were resized to 160 × 160 pixels.

Split	Images	Share
Training	958	70.0%
Validation	205	15.0%
Test	206	15.0%
Total	**1,369**	**100%**

Class counts: n0 131; n1 139; n2 137; n3 152; n4 131; n5 141; n6 132; n7 142; n8 133; n9 131.

3. Model and approach

The backbone was Keras MobileNetV2 with ImageNet weights and include_top=False. Inputs were rescaled from [0,255] to [-1,1]. The custom head used GlobalAveragePooling2D, Dropout, and a ten-unit Dense softmax output. Feature extraction froze the base and trained the head. For fine-tuning, the selected top layers were unfrozen while BatchNormalization layers stayed frozen; training continued with a lower learning rate. Augmentation, when enabled, applied random horizontal flips, rotations, and zooms.

4. Experiment tuning results

Each run used five feature-extraction epochs and, where configured, three fine-tuning epochs. Accuracy values below are the notebook's final-epoch training and validation accuracies.

Exp	LR head	LR fine	Batch	Unfreeze	Dropout	Aug.	Train	Val.
1	.001	.0001	32	0	.2	No	99.27%	97.56%
2	.001	.0001	32	20	.2	Yes	97.49%	98.54%
3	.0005	.00005	32	30	.3	Yes	96.45%	97.56%
4	.001	.00001	16	50	.3	Yes	98.02%	98.05%
5	.0003	.00003	32	20	.5	Yes	92.38%	97.07%
6	.001	.0001	64	40	.2	Yes	97.39%	98.05%

5. Best configuration and observations

Experiment 2 had the highest recorded validation accuracy (98.54%). It used MobileNetV2, 20 top layers unfrozen, head learning rate 0.001, fine-tuning learning rate 0.0001, batch size 32, dropout 0.2, augmentation enabled, five head-training epochs, and three fine-tuning epochs. Experiment 1 without augmentation reached 97.56% validation accuracy. Experiment 5, with dropout 0.5, had the lowest final training and validation accuracies (92.38% and 97.07%).

Experiment 2's final training accuracy (97.49%) was slightly below its validation accuracy (98.54%); this does not indicate an obvious final-epoch train–validation gap. Since multiple settings differ across runs, the results do not isolate the causal effect of any single hyperparameter.

6. Training curves

The Colab notebook contains the plotting code for training/validation accuracy and loss for both stages of the selected run. The plot images were not available in the notebook content extracted for this report. Insert the actual feature-extraction and fine-tuning plots from Colab here before submission.

7. Final test-set evaluation

The notebook reserves 206 images for the test set. A recorded final test evaluation was not present in the saved outputs available for this report, so test accuracy is not stated. Run the final evaluation cell once after choosing Experiment 2 and enter the resulting accuracy here. Do not use the test result to tune the model.

8. Sample predictions

The assignment requires a 5 × 5 grid of at least 25 held-out test images with true and predicted labels. The generated grid was not present in the extracted notebook outputs. Insert the actual prediction-grid output from Colab here before submission.

9. Summary

The six runs achieved final validation accuracies from 97.07% to 98.54%. Experiment 2 was the strongest observed configuration, using augmentation and limited fine-tuning. Higher dropout in Experiment 5 coincided with weaker results in that configuration. Because the experiments changed several settings at once, these observations are descriptive rather than causal. The report assumes the original training and validation folders were pooled for a new stratified 70/15/15 split, as implemented in the notebook.

References

Kaggle dataset: 10 Monkey Species
Keras MobileNetV2: Keras Applications
Colab notebook: public notebook